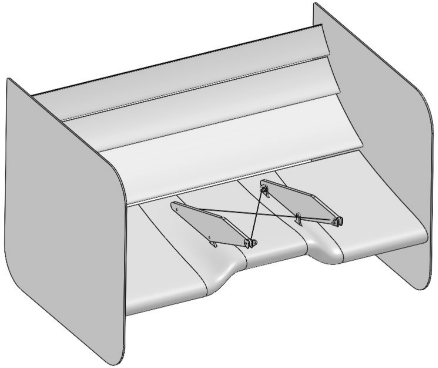
CARNEGIE MELLON RACING



Rear Wing PPDR

System Lead: Malik Choset
Captain: Vincent Chen

DESIGN REVIEW ATTENDANCE

- Invites:
  - 27x Aero:
    - Vincent Chen
    - Anika Zhou
    - Sheri Zhang
    - Yuchi Lu
    - Vyju Vasudevan
    - Ben Reske
  - Aero Alumni:
    - Tommy Gordon

- Invites:
  - 27x Leadership:
    - Nate

- PPDR attendance: (on day of review)
  - Leads + Vincent
  - Tommy
  -

AGENDA

- Aerodynamics
  - Goals
  - Bounding Box & RTML Compliance
  - Airfoil Selection (2D)
  - 3 Elem vs 4 Elem
  - Airfoil Validation (3D)
  - DRS
- Structures
  - Tabs
  - Cables

DESIGN OBJECTIVES

Overall Goals

- Adjustability
  - Maximize DF in Autocross & Skidpad
    - Current bottleneck for package CL potential
  - Minimize Drag in Endurance and Accel
- Multi-Element Airfoil Selection
  - Move away from Selig (high camber) airfoils to aid more stable performance
  - Minimize drag in open position
  - Maximize DF in closed position
- Eliminate slop in mounting
  - Cables w/ tensioning mechanism

System Goals

Target	26x Performance	Plan to improve
Mass	5.62 kg	With DRS Removed at most 10% increase in mass
Cl	__	
Cd	__	

- Skin Mass
  - Current Surface Area: 1.73 m^2 → 1.8 kg
  - 26x Surface Area: 1.6 m^2 → 1.56 kg

Historic Problems

- Slop in mounting system
- Takes a long time to take on and off
- Flaps failing in closed position
- RWG Cl bottlenecks FWG Cl
- Leading Edge issues created during MFG Process
- DRS lol

Constraints: Rules

EV.6.2.5 Mounting location of each Ready to Move Light must:

a. Be near the Main Hoop near the highest point of the vehicle

b. Be inside the **Rollover Protection Envelope F.1.13**

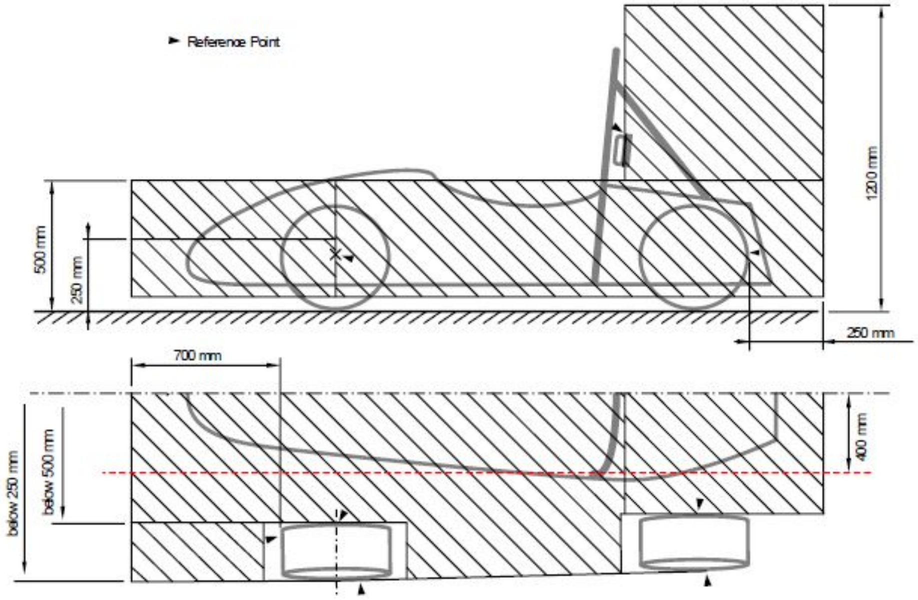
c. Be no lower than 150 mm from the highest point of the Main Hoop

d. Not let the driver's helmet touch it

e. Be visible from 1300 mm vertically from ground level, inside a 2000 mm horizontal radius from the light, with Bodywork and Aerodynamic Devices in place

Constraints: Rules

- T.7: Bounding Boxes



Constraints: Loads

Physical Loading:

- Aero Load + 5g Bump
  - Aero Load from 26x MSHD 3-Elem (17.5m/s): 284
  - Projected Mass (1.1x 26x RWG): 7kg
  - Total: 627N

Constraints: Clearances

Spatial Constraints

- Aero BB
- BB Created by RTML Visibility
- Roll Hoop

Constraints: Monocoque Mounting

Design consideration for monocoque

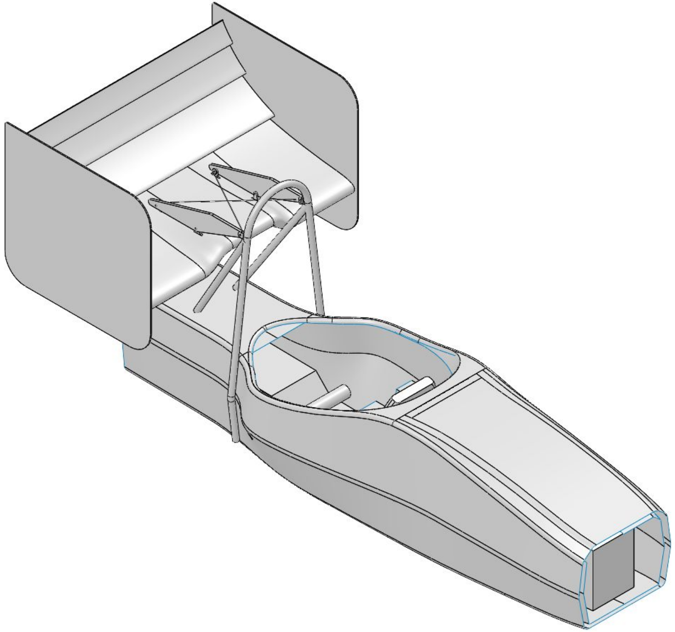
- Tabs onto Roll Hoop (will cover later)

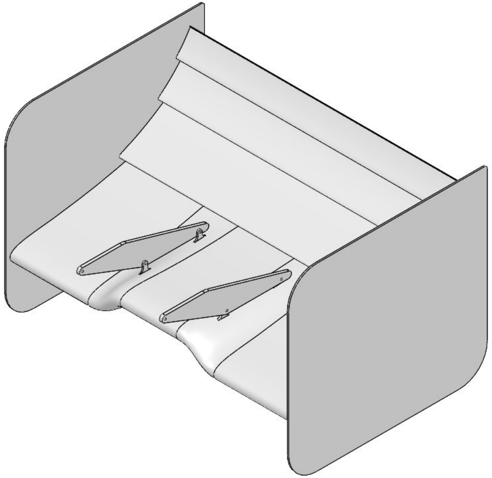
Constraints: Electrical Systems

- DRS - Will discuss after PPDR

SYSTEM ARCHITECTURE

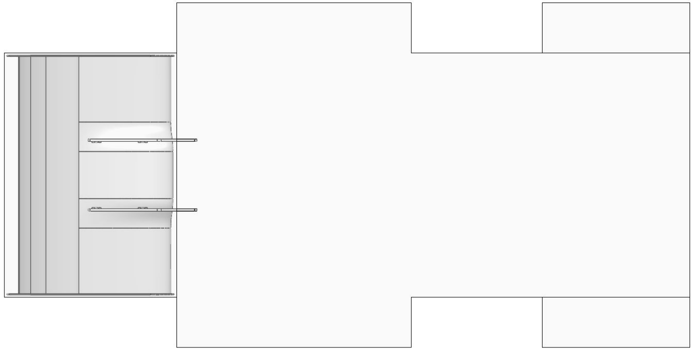
System Overview





Bounding Box

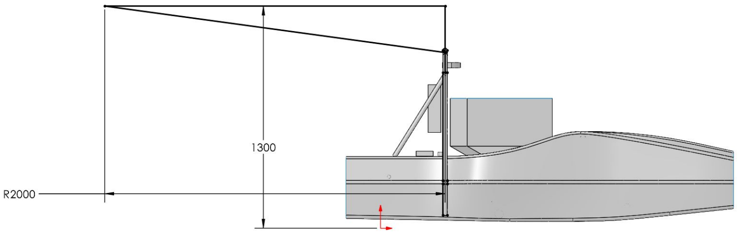
Side View

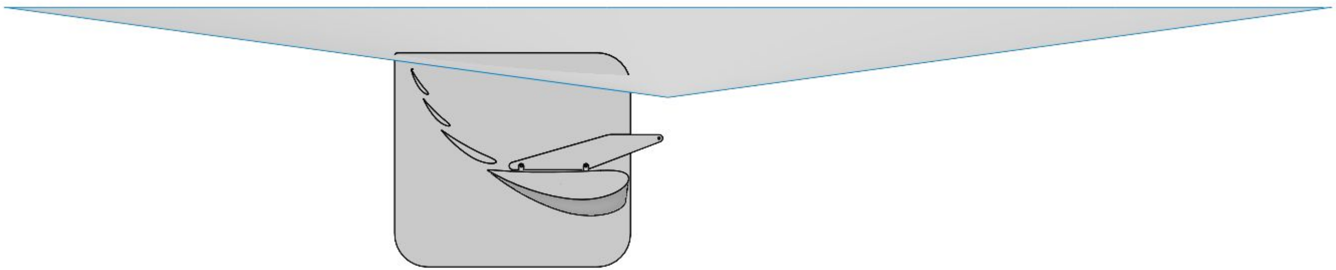


Top View

*Closest nominal distance to a BB surface is 10mm

RTML Bounding Box





* The upper part of the endplates that are not RTML BB Compliant will be resolved in an endplate optimization template

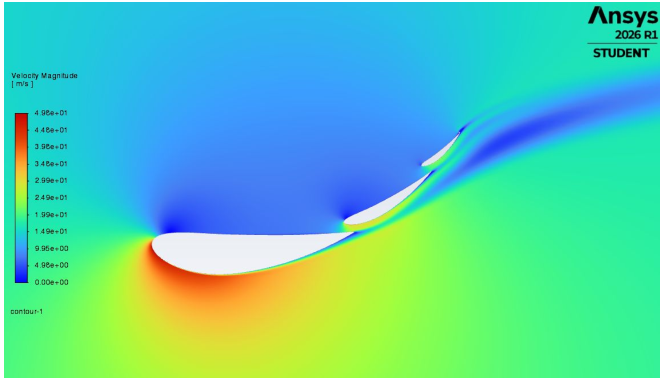
Airfoil Selection

- Goal
  - Use less cambered airfoils to support more stable performance
    - Using less camber should increase pitch stability in the multi-element chain because on each airfoil there is less of an adverse pressure gradient to overcome
  - Chose airfoils that would minimize drag in an open position while maintaining adequate Cl in their closed position
- Methodology
  - Chose airfoils based off of results in 2D Sims
  - Validate choices in 3D Geo by comparing to 26x 3-Elem Wing

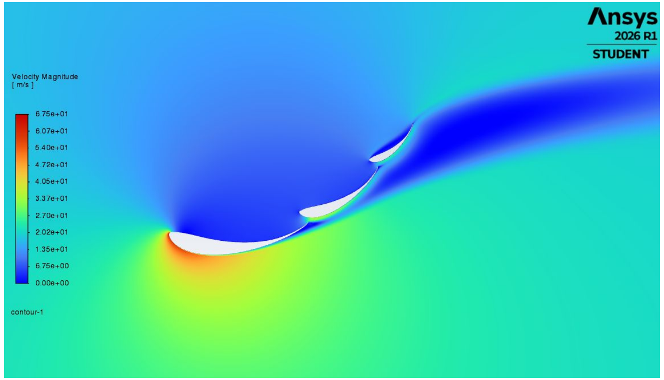
Airfoil Selection cont.

- Started with a NACA 4420 - 4412 - 4412 chain
- Optimized foils by increasing camber & thickness until flow separation occurred
- Resulting in NACA 8420 - 7412 - 7412 chain

NACA Chain (Cl: 2.81)

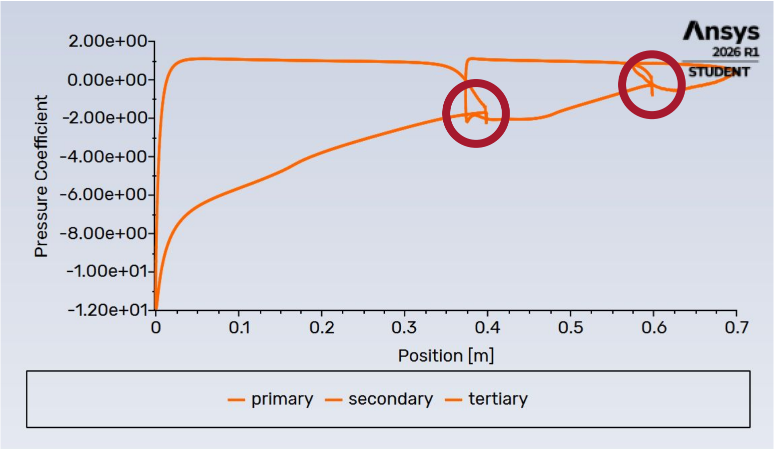


MSHD Chain (Cl: 2.45)

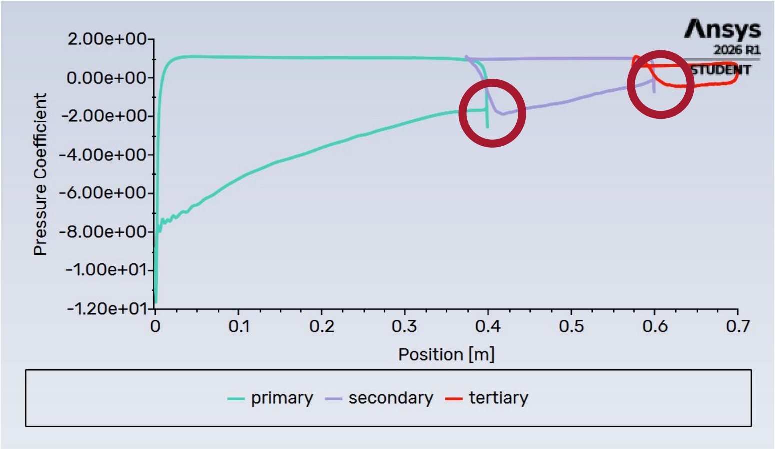


Airfoil Selection cont.

NACA Chain



MSHD Chain



Less aggressive pressure recoveries shown in circled areas → promotes better TE recovery

3 vs 4 elem study

- Simmed simple 3 and 4 elem RWG's (2D extrusions w/ endplates)
- Takeaways
  - Same Cd in open positions, therefore closed position will drive results
  - 3 vs 4 elem decision will be based on closed position Cl vs Mass

	3 Elem Closed	3 Elem Open	4 Elem Closed	4 Elem Open
RWG Cl:	0.84	0.11	0.95	0.13
RWG Cd:	0.38	0.03	0.43	0.03

Grey indicates metric we care about in that state for rear wing

3 vs 4 elem study mass estimation

4 Elem			**3 Elem**	
Foil Area (m^2)	1.58		Foil Area (m^2)	1.60
Endplate Area (m^2)	0.401		Endplate Area (m^2)	0.38
Skin Mass (kg)	1.66		Skin Mass (kg)	1.68
Endplate Mass of one (kg)	0.72		Endplate Mass of one (kg)	0.70
Primary Rib Mass (g)	97.0		Primary Rib Mass (g)	98.1
Secondary Rib Mass (g)	16.3		Secondary Rib Mass (g)	22.8
Tertiary Rib Mass (g)	5.74		Tertiary Rib Mass (g)	13.9
Quaternary Rib Mass (g)	3.74			
Total Mass (kg)	**3.72**		Total Mass (kg)	**3.73**

- Skin GSM Value
  - 1050 g/m^2
- Endplate GSM Value
  - 1800 g/m^2
- Ribs are solid, 2mm thick, 6061
  - Mass estimate from SW
- Mass underestimation
  - Hardware
  - Lack of Swan necks
  - No DRS
  - No epoxy

3 vs 4 elem study cont.

	3 Elem Closed	4 Elem Closed
RWG Cl:	0.84	0.95
RWG Mass (kg):	3.73	3.72
RWG Cl/Mass:	0.225	**0.255**

Reasons for choosing 4 Elem

- Higher Cl/Mass in Closed Config
- RWG is Cl bottleneck
    - 4-Elem better reduces the bottleneck compared to 3-Elem

Airfoil Validation (in progress)

26x MSHD Wing

	Closed Config	Open Config	Δ	% Diff
RWG Cl:	1.51	0.55	0.96	63.6%
RWG Cd:	0.65	0.18	0.47	72.31%

27x 4-elem wing

	Closed Config	Open Config	Δ	% Diff
RWG Cl:	1.22	0.29	.93	76.2%
RWG Cd:	0.54	.08	.46	85.1%

Airfoil Validation (in progress)

26x MSHD Wing

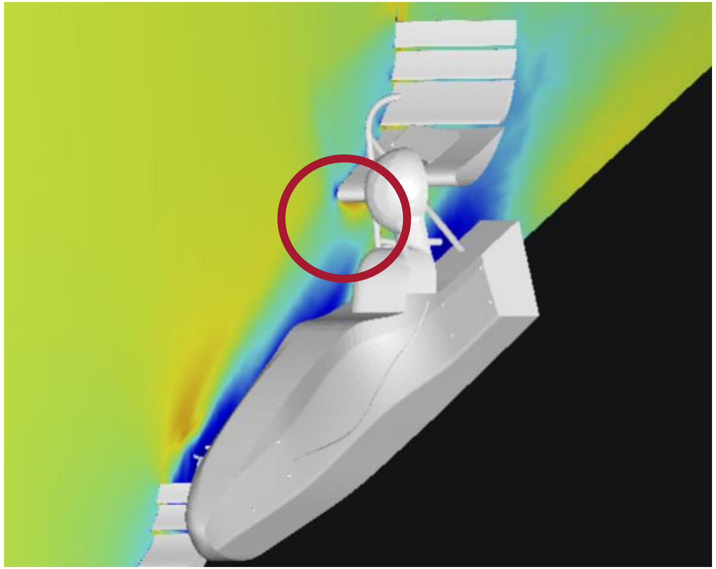
	Closed Config	Open Config
RWG Lift:	284 N	103.2 N
RWG Drag:	124 N	34.4 N

27x 4-elem wing

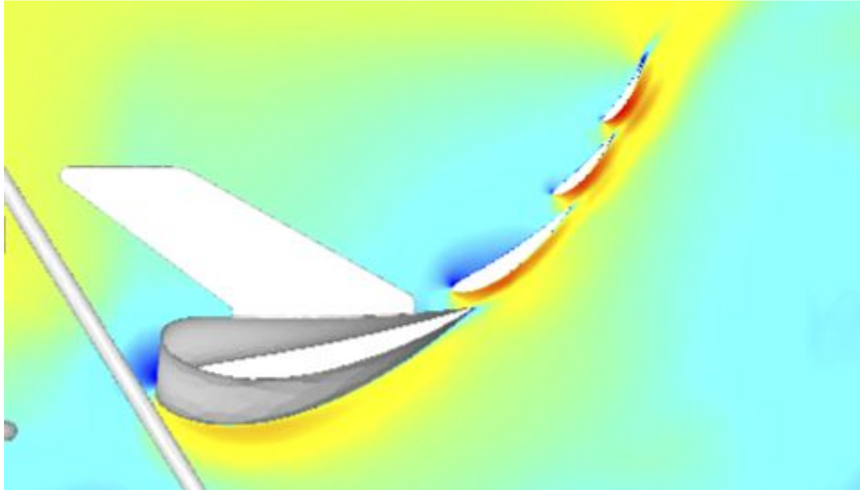
	Closed Config	Open Config
RWG Lift:	230 N	53.8 N
RWG Drag:	102.2 N	14.94 N

Airfoil Validation Gate (Before RTML Discussion)

Unoptimized main element clears FWG Wake

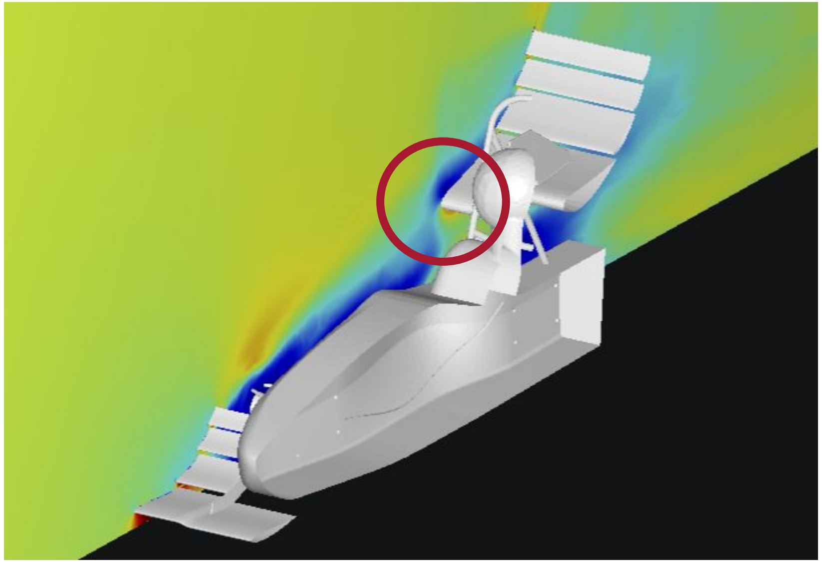


Doesn’t show signs of flow separation

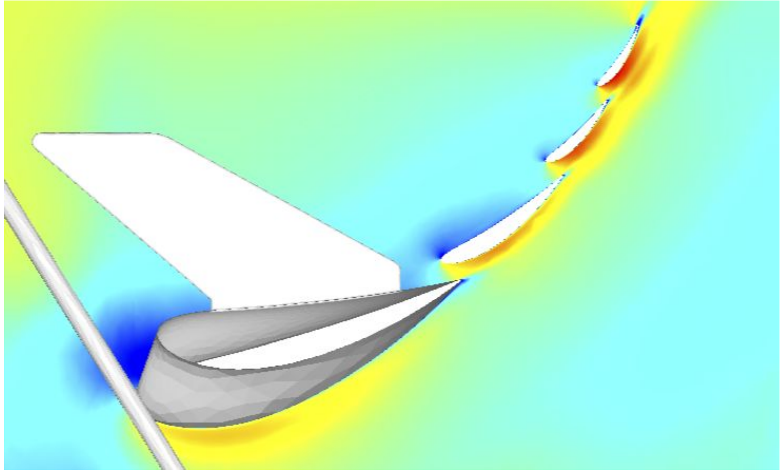


Airfoil Validation Gate (Before RTML Discussion)

Main element pushed harder comes into FWG Wake



Still doesn't show signs of flow separation

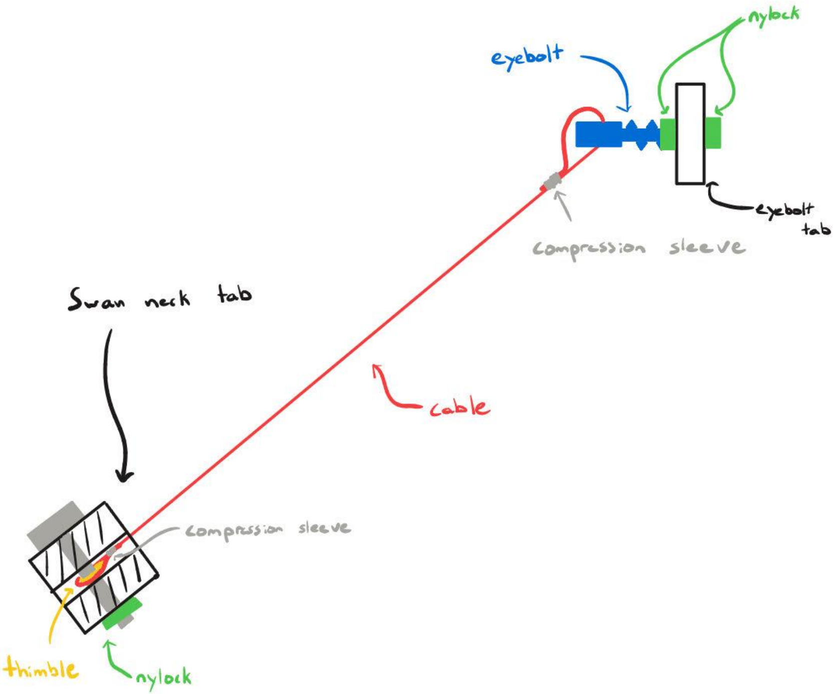


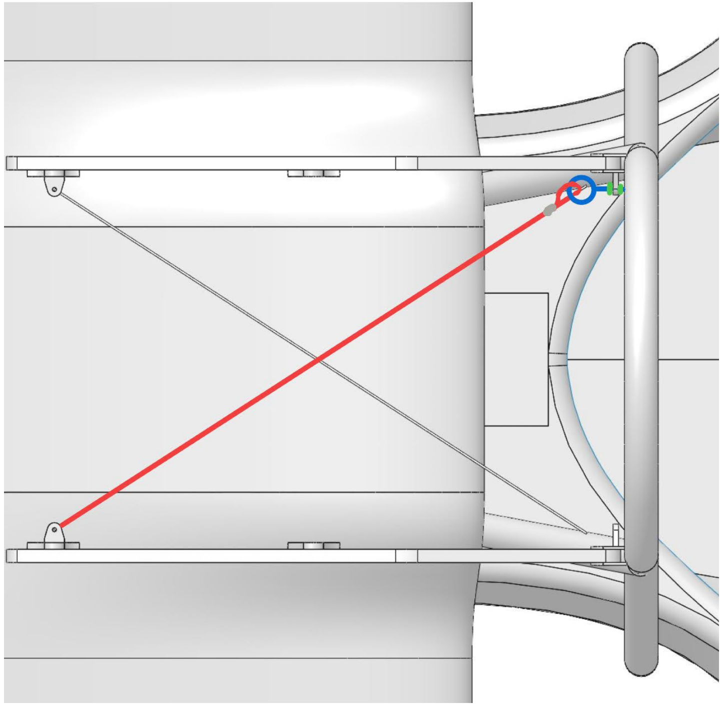
Airfoil Validation Gate (After RTML Discussion)

- FWG Wake no longer kills RWG performance
- Haven’t had time to run any studies to optimize RWG since RTML discussion yesterday
- Steps to gain Cl will be discussed on steps leading to PDR
- The same RWG gained 0.25 Cl after being raised to top of RTML BB

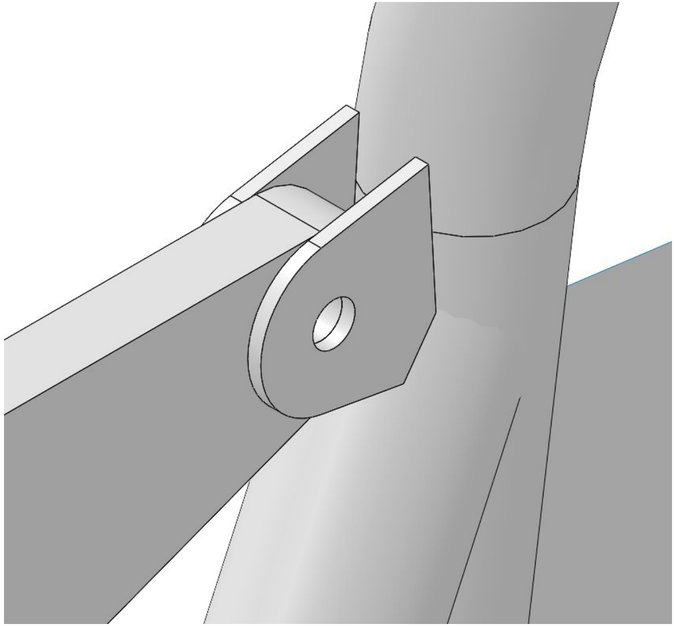
Mounting

Tensioning Mechanism

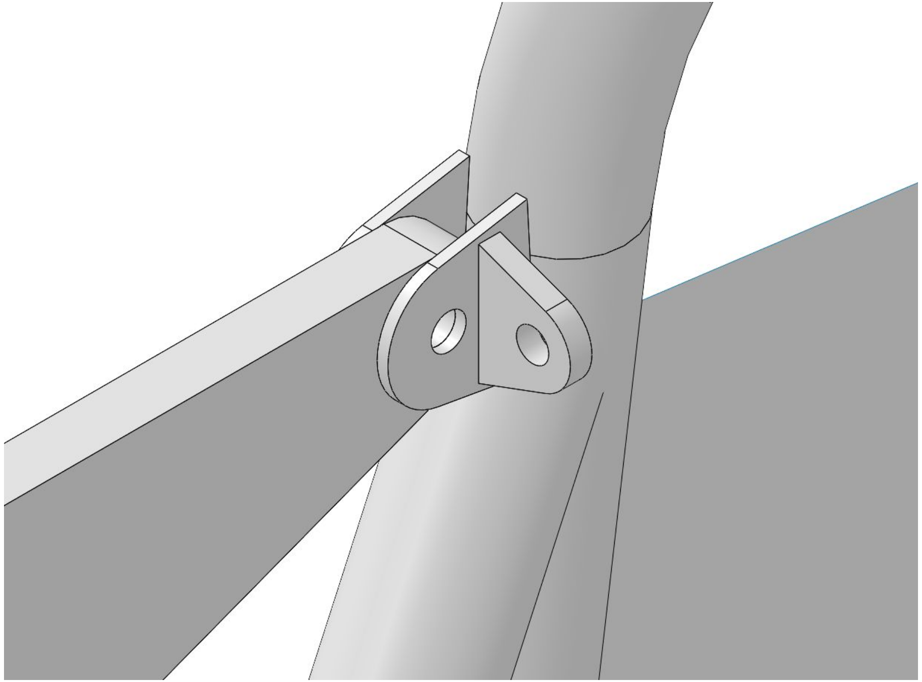




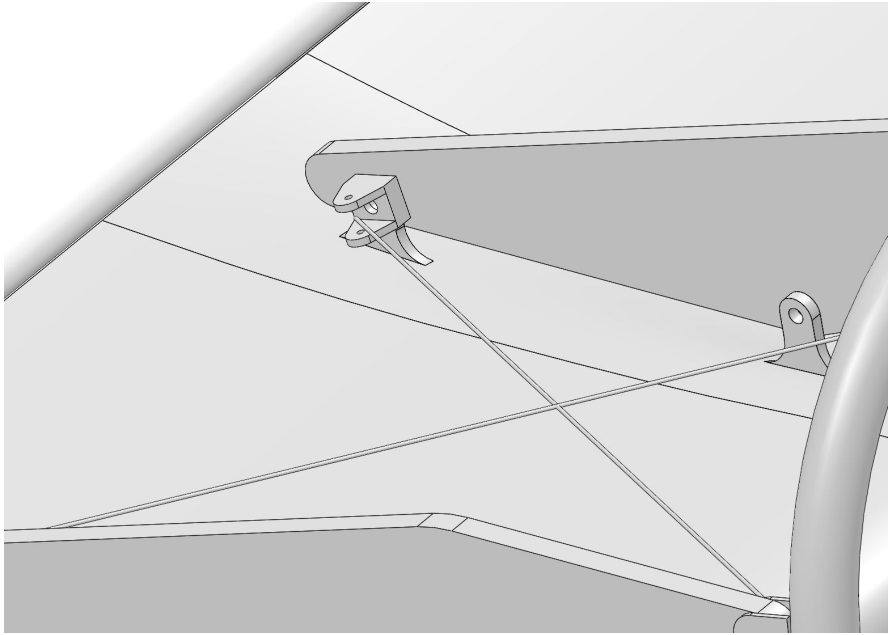
MRH Tabs

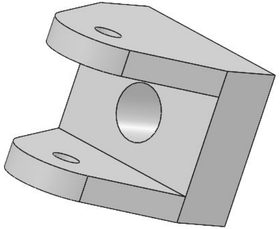


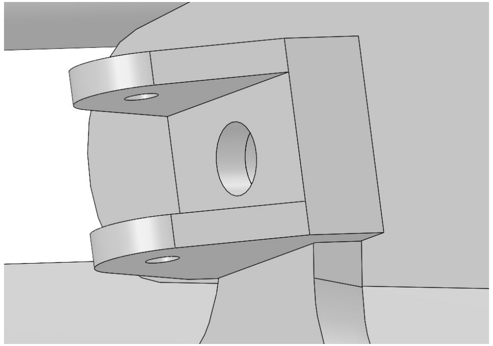
Eyebolt Tabs



Swan neck Tabs







IMPLEMENTATION & LOGISTICS

Timeline and Deliverables for PDR

- Aerodynamics
  - Main Elem optimization
  - Chord Length & Flap AoA optimization
  - Gap spacing optimization
  - Endplate optimization
  - Pitch & Yaw Studies
- Structures
  - Rib & Spar
  - Mounting mechanism fleshed out
  - Bottom Struts
  - Endplate Stiffness
  - Male Plug Mold testing

Concerns

- After optimization RWG might not get to Cl of 26x 3-elem (1.5)
- Eye Bolts bending
- Swan necks are tiny
- Endplate stiffness needs to be evaluated